

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination – 2024 Course: B.Tech Branch : Food Engineering & technology/ Food Tech & Managemnet Semester : VI Subject Code & Name: BTFT602 Chemical Reaction Engineering Max Marks: 60 Date 20/12/2024 Duration: 3 Hr.				
Instructions to the Students: 1. Each question carries 12 marks. 2. Question No. 1 will be compulsory and include objective-type questions. 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6. 4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 5. Use of non-programmable scientific calculators is allowed. 6. Assume suitable data wherever necessary and mention it clearly.				
			(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)			12
1	What is the chemical reaction rate?			1
	a. What is the chemical reaction rate?	b) Chemical reaction rate is a physical property	c) Chemical reaction rate is an intensive property	d) Chemical reaction rate is an extensive property
2	Which of the following deals with the study of rates at which chemical reactions occur and the effect of various parameters on the rate?			1
	a) Fluid Mechanics	b) Chemical Technology	c) Chemical Sciences	d) Chemical Kinetics
3	3. If $\tau=13$, average conversion for a particle B of constant size in a mixed flow reactor for chemical reaction controlling is?			1
	a) 0.75	b) 0.92	c) 0.76	d) 0.98

4	Which of the following represents heterogeneous catalytic reaction?					1
	a) Reduction of iron ore	b) Ammonia synthesis	c) Burning of coal	d) Roasting of ores		
5	Which of the following is the most suitable reactor for the pharmaceutical industry?					1
	a) PBR	b) MFR	c) PFR	d) Batch reactor		
6	If the conversion of a first-order liquid phase reaction occurring in a CSTR is 75%, molar feed rate is 5 mol/min, the rate of the reaction is 5 mol/litre.min then the volume of the reactor (in litre) is?					1
	a) 0.33	b) 0.4	c) 0.75	d) 0.5		
7	12. The rate constant of a first order reaction depends on the _____					1
	a) Temperature	b) Concentration	c) Time	d) Pressure		
8	Which of the following is an example of autocatalytic reaction?					1
	a) Oxidation of copper	b) Ammonia formation reaction	c) Hydrogen peroxide decomposition reaction	d) Reaction of permanganate with oxalic acid		
9	Which of the following reactor arrangements causes fast deactivation?					1
	a) Fluidised bed reactor	b) Batch for fluid and solid	c) Plug flow for fluid	d) Mixed flow for fluid		
10	If $\tau = 5$ s, first order rate constant, $k = 0.25 \text{ sec}^{-1}$ and the number of tanks, N is 5, then the conversion is?					1
	a) 87.45%	b) 33%	c) 75%	d) 67.2%		
11	The catalyst in a first-order chemical reaction changes the _____					1

	a) Activation energy	b) Equilibrium constant	c) Heat of reaction	d) Heat of formation of the product		
12	The rate of reaction is rapid for which of the following reactions?					1
	a) Gas reactions in porous catalysts	b) Reactions in rocket engines	c) Cellular reactions	d) Reactions in coal furnaces		
Q. 2	Solve the following.					12
A)	Give a brief account on chemical reactor. 6					6
B)	Different types of reactors & write down the characteristics of reactors					6
Q. 3	Solve the following.					12
A)	What is the difference between selectivity and yield?					6
B)	Give a brief review of the kinetics of homogeneous reactions?					6
Q. 4	Solve Any Two of the following.					12
A)	Explain in detail one parameter model.					6
B)	Give a brief review of the kinetics of homogeneous reactions?					6
C)	Differentiate Elementary vs Non elementary Reaction?					6
Q. 5	Solve Any Two of the following.					12
A)	What is a pseudo first order reaction? Give an Example					6
B)	Explain temperature dependents of rate equation by various theories?					6
C)	Distinguish between homogeneous and heterogeneous reactions					6

Q. 6	Solve Any Two of the following.		12
A)	Differentiate Elementary vs Non elementary Reaction?		6
B)	Write down the performance equation for recycle reactor?		6
C)	Explain temperature dependents of rate equation by various theories?		6
	*** End ***		